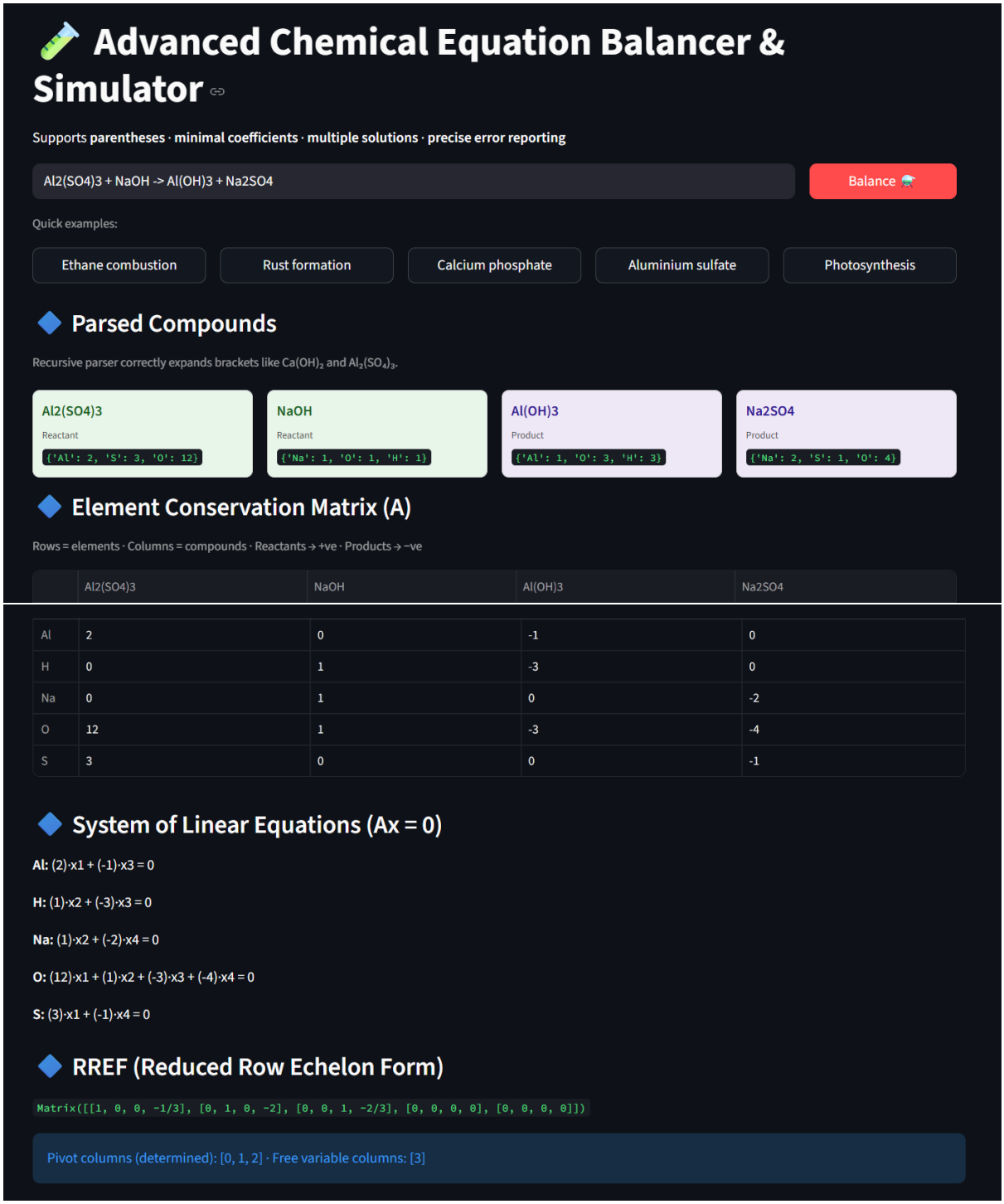
Screenshots :



Rank & Nullity

Rank	Nullity	Compounds
3	1	4

Null Space

```
[
  0 : "Matrix([
      [1/3],
      [  2],
      [2/3],
      [  1]])"
]
```

1 basis vector(s). Each encodes one independent coefficient ratio.

LCM Scale → GCD Reduce → Minimal Coefficients

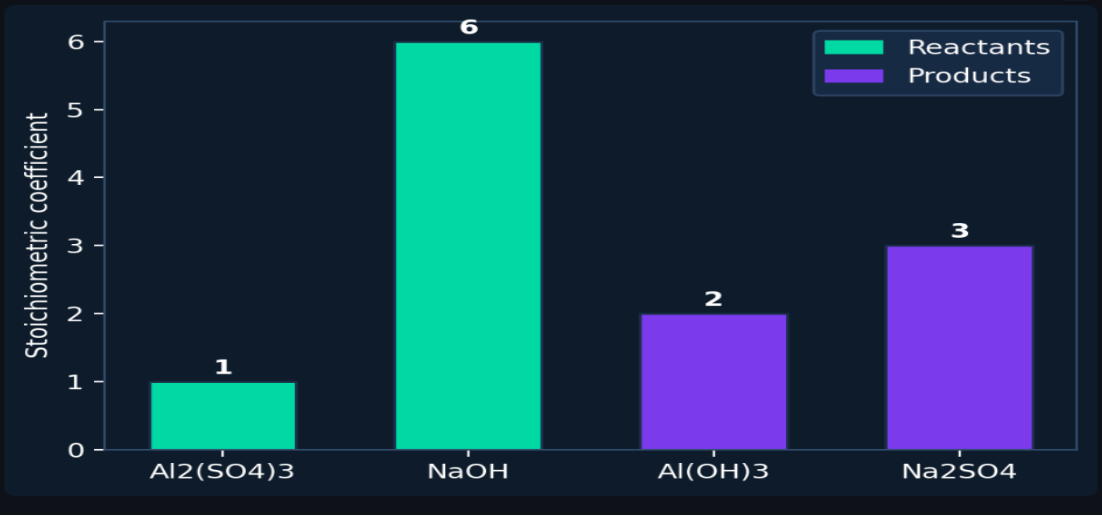
Compound	Null vector	× LCM(3)	÷ GCD(1)
Al2(SO4)3	1/3	1	1
NaOH	2	6	6
Al(OH)3	2/3	2	2
Na2SO4	1	3	3

✅ Balanced Equation

Al2(SO4)3 + 6NaOH → 2Al(OH)3 + 3Na2SO4

✓ Numerically verified: atom counts match on both sides (Ax = 0 confirmed).

Mole Ratio Visualisation



Limiting Reagent Simulator

Enter moles of each reactant. App shows which runs out first and how much product forms.

Moles of Al2(SO4)3

1.00 − +

Moles of NaOH

1.00 − +

Limiting reagent: `NaOH`

Ratio vector (available ÷ coefficient): [1.0, 0.1667]

Minimum ratio = 0.1667 → `NaOH` exhausted first

Products formed:

→ `Al(OH)3` : **0.3333** moles

→ `Na2SO4` : **0.5000** moles

Excess reactants remaining:

→ `Al2(SO4)3` : **0.8333** moles unused